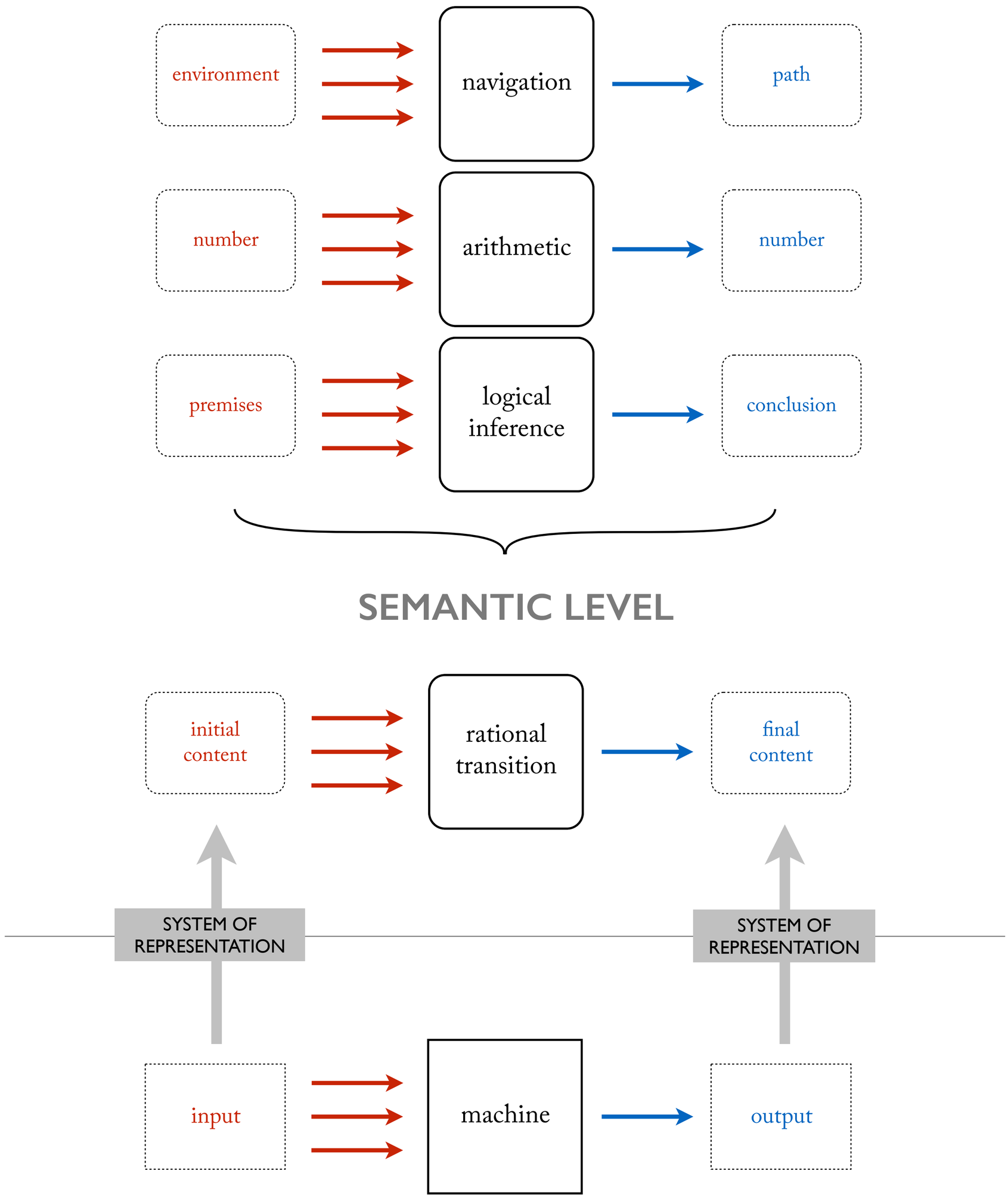
4. Functions

The Structure of Computation

So far we've described computational information processing in terms of rational transitions between informational content. Now we'll use the mathematical concept of a function to formalize this idea.

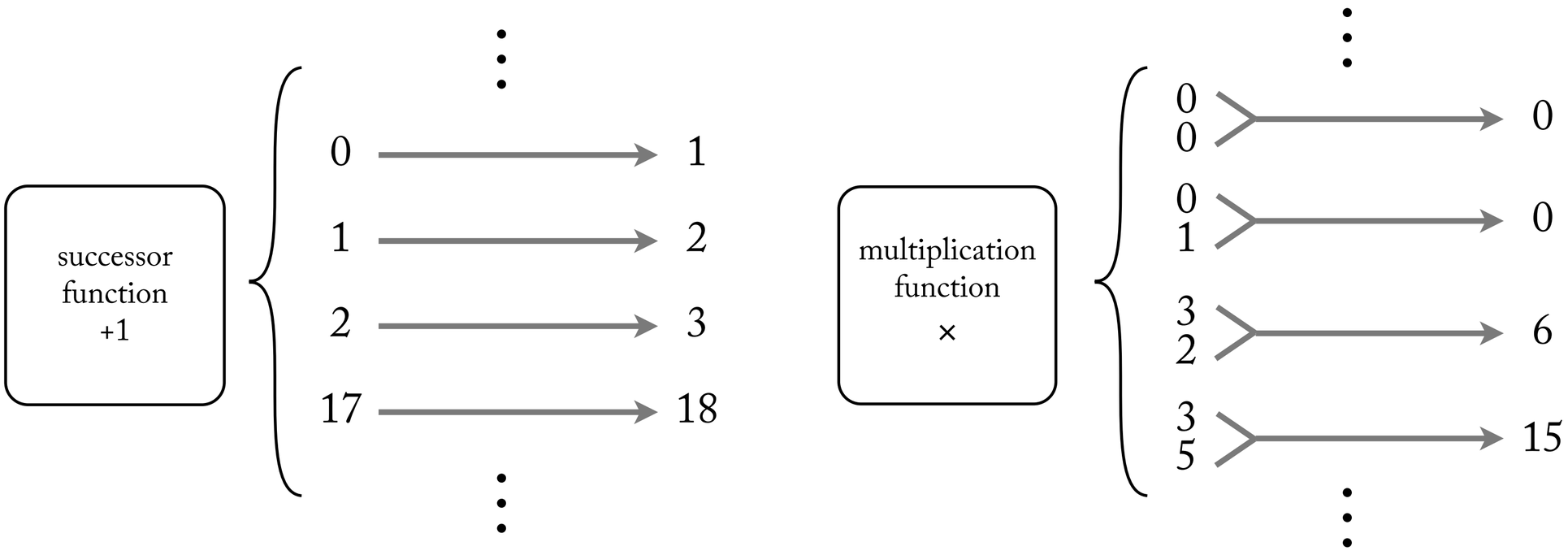
Officially, we'll say that a computer **computes a function** when it performs a given rational transition.

Rational transitions as computable functions…



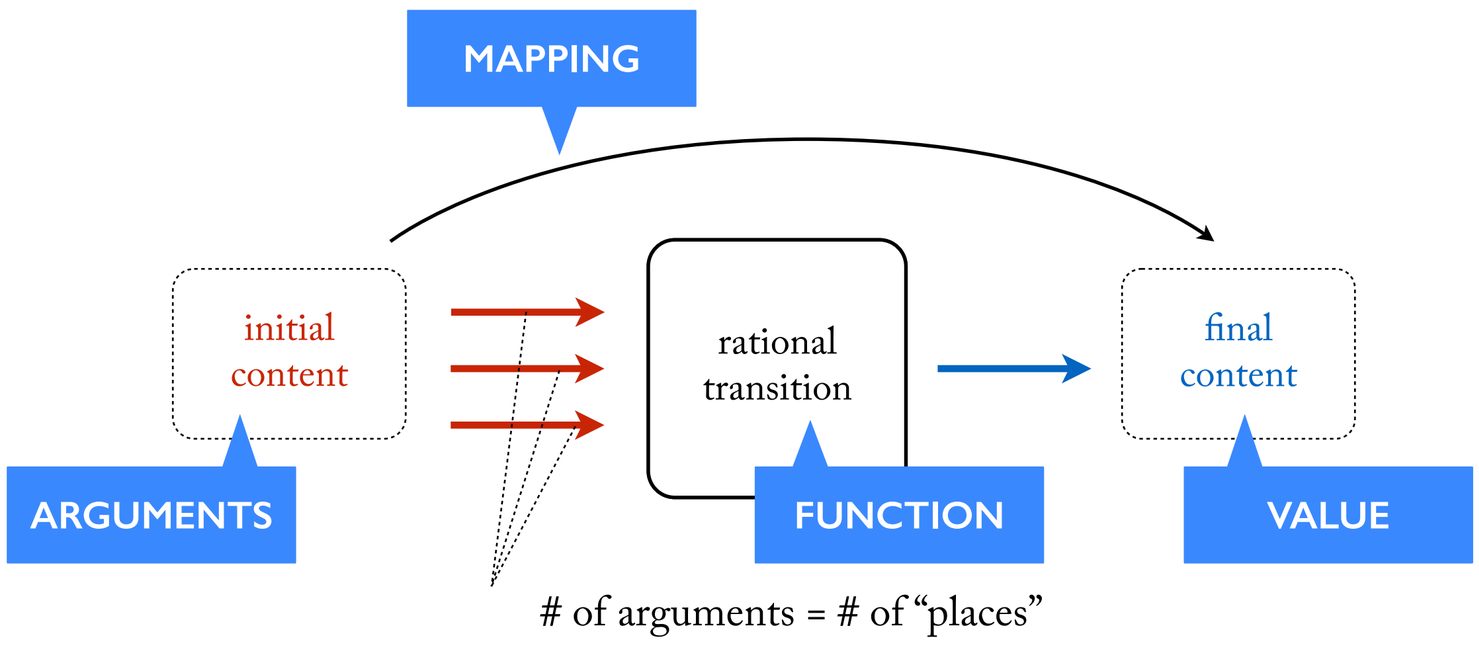
Numerical Functions

Familiar functions correspond to relations between numbers. Examples include *successor*, *addition*, *multiplication*, and so on.



If you think of numbers as abstract objects in Plato's heaven, then functions are abstract and infinite relations *between* those numbers.

The Structure of Functions



Functions are **deterministic**. Given the same arguments, they always return the same value.

Functions can have **many** arguments, but only **one** value.

Multiple arguments go in different **argument places**. What is a function for which argument place matters?

the addition function:

$$3 + 4 = 7$$

argument 1 argument 2 value

Functions: Terminology

How to **talk** about functions…

The function **takes X** (as an argument) and **returns Y** (as value).

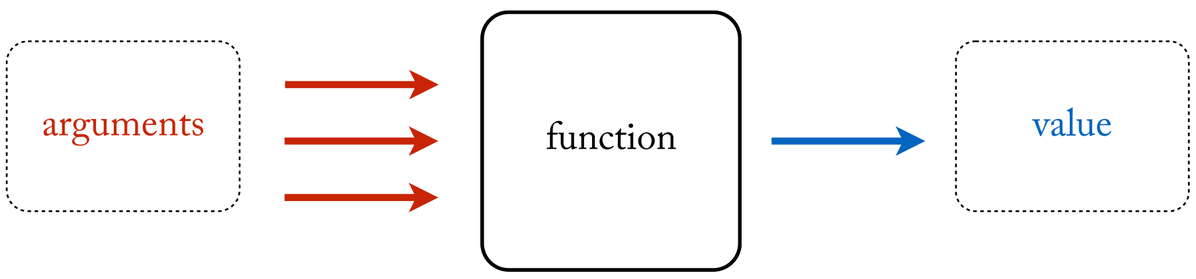
The function **applied to X, returns Y**.

The function **maps X to Y.**

Functions: Notation

How to **write** about functions…

Prefix notation: $f(x, y) = z$



Infix notation: $x \, f \, y = z$

Prefix notation puts the function symbol on the left side.

"add( )" means the addition function
"add(x, y)" means the addition function applied to arguments x and y

For example:
"add(3,4)" means the addition function applied to the numbers three and four
which is... the number seven

Which is why it is true to say: add(3,4) = 7

Infix notation (Grade School Notation) puts the function symbol in the middle.

"+" means the addition function
"x+y" means the addition function applied to arguments x and y

For example:
"3+4" means the addition function applied to the numbers three and four
which is... the number seven

Which is why it is true to say: 3+4 = 7

Functions: Composition

Functions can be combined together. This is called the **composition** of functions. Feel free to mix the two styles of notation together as you find convenient.

$(3 + 4) \times 2 = 14$

multiply(add(3,4), 2) = 14

add(3,4) × 2 = 14

These statements are all equivalent, and appropriate use of notation.

Defining a Function

The value of a function may be **undefined** for some arguments.

divide(3,0) = undefined *square.root*(-4) = undefined

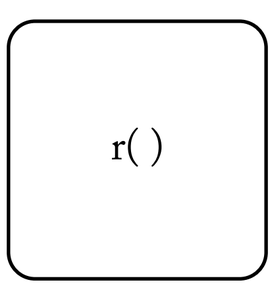
Most functions are defined for **infinitely many** arguments. Some functions are defined only for **finitely many** arguments. Call these **finite functions**.

To define an **infinite function** you typically have to **use an equation.**

Example 1
Let s( ) be the 1-place function which satisfies the following condition:
For any number x: $s(x) = x + 1$

Example 2
Let q( ) be the 2-place function which satisfies the following condition:
For any numbers x, y: $q(x,y) = x + (y \times 2)$

To define a **finite function** you typically have to **use a table.** (See below for an exception.) By convention, the function is undefined for the unlisted argument pairs.



=

argument 1	argument 2	value
0	0	15
0	1	3
1	0	0
1	1	1
1	2	2

The Identity of Functions

A function is defined by its arguments and its values *only*.
It doesn't matter *how* these are specified.

Example 1

Let f( ) be the function:
for any x: $f(x) = x^2$

Let g( ) be the function:
for any x: $g(x) = x \times x$

It follows that f( ) = g( ).

Example 2

Let h( ) be the function:

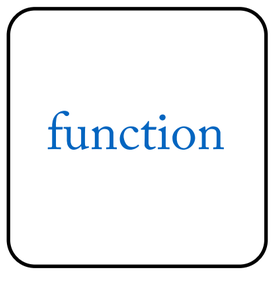
argument	value
1	2
2	3
3	4

Let j( ) be the function:
for any x:
$j(x) = x + 1$ if $1 \leq x \leq 3$:
$j(x)$ = undefined otherwise

It follows that h( ) = j( )

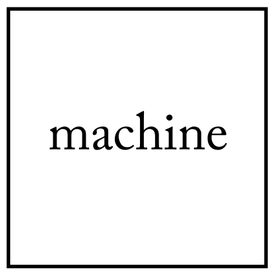
Functions vs Machines

In this class it's critical to keep your concept of **function** and your concept of **machine** clearly separated. (Even if, in some of the readings, authors sometimes run them together!)



A function is…

- An abstract entity, which often relates numbers.
- Has no temporal or spatial location.
- Up to infinite in size.
- "Instantaneously" relates one number to another.
- Requires no "processing" to do this.



A machine is…

- A concrete device, which processes symbols.
- Has a temporal and spatial location.
- Only finite in size.
- Takes time to take an input and produce an output.
- Requires processing to do this.

The Metaphysics of Functions

Mathematical functions are abstract infinite relations between numbers. There is nothing more to a function than the numbers it relates.

You can think of a mathematical function as an infinite "list" of numbers.

Other kinds of functions can relate other kinds of objects (sets, people, places, etc.). But the functions themselves are always abstract relations.

argument 1	argument 2	value
0	0	0
0	1	0
1	0	0
1	1	1
1	2	2
2	0	0
2	1	2
2	2	4
3	0	0